

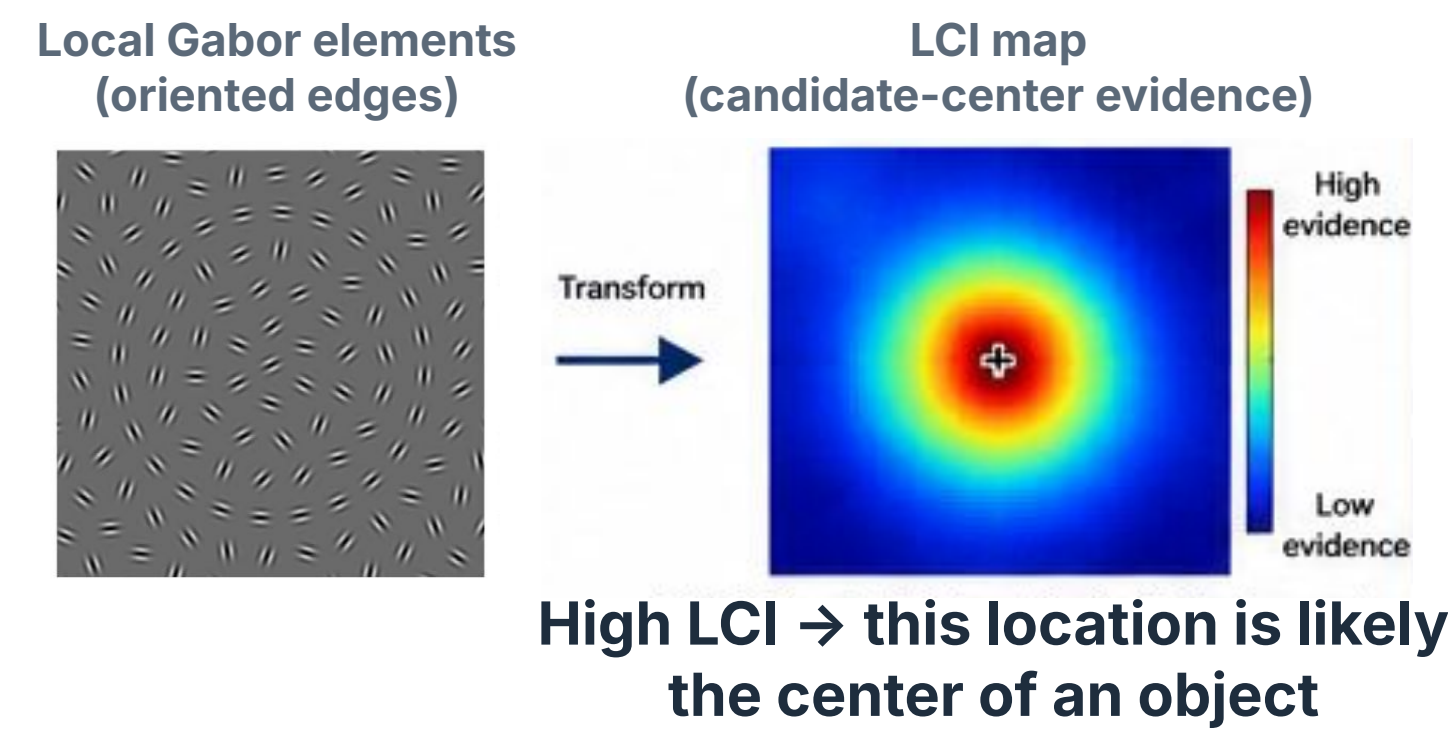
Human gaze preferentially samples local concentric structure during visual search

Saemi Jung¹, Juhyung Ryu¹, Sang-Hun Lee^{1*} | ¹ Department of Brain and Cognitive Sciences, Seoul National University, Seoul, Republic of Korea

1 BACKGROUND & HYPOTHESIS

What is Concentricity?

How consistently local orientations support the same candidate center



Previous Studies

1 FOVEATED VISION

Acuity peaks at the fovea and falls with eccentricity.

Geisler et al. (2006)

2 ACTIVE VISUAL SAMPLING

Eye movements relocate the fovea to reduce uncertainty.

Najemnik & Geisler (2005)

3 OBJECT-CENTERED LANDING

Object-entering fixations land near object centers.

Nuthmann & Henderson (2010)

4 NATURAL EDGE RELATIONS

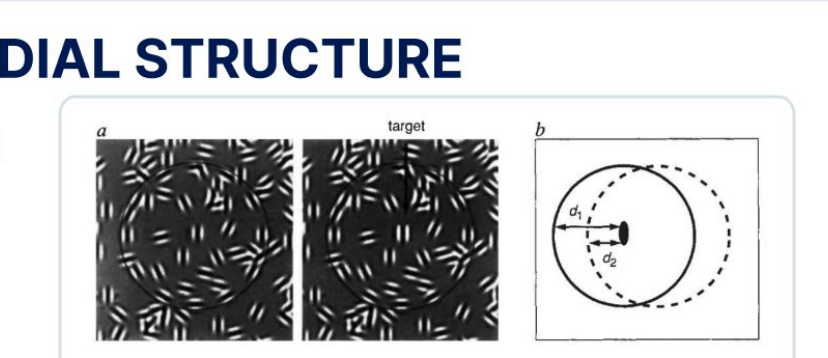
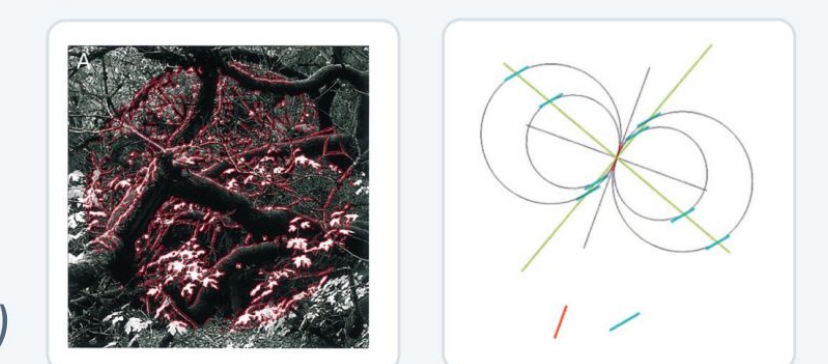
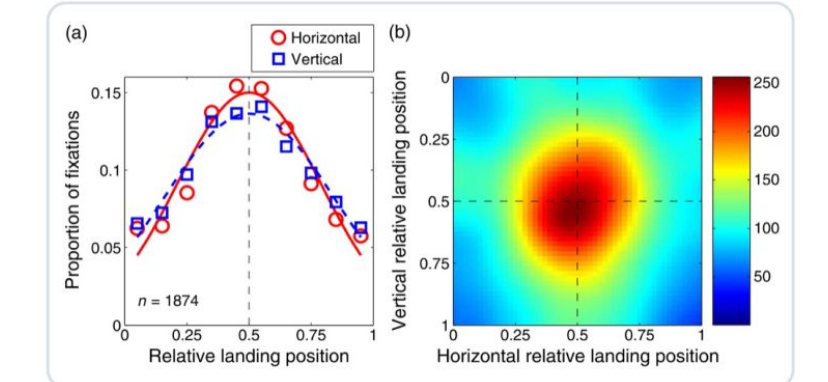
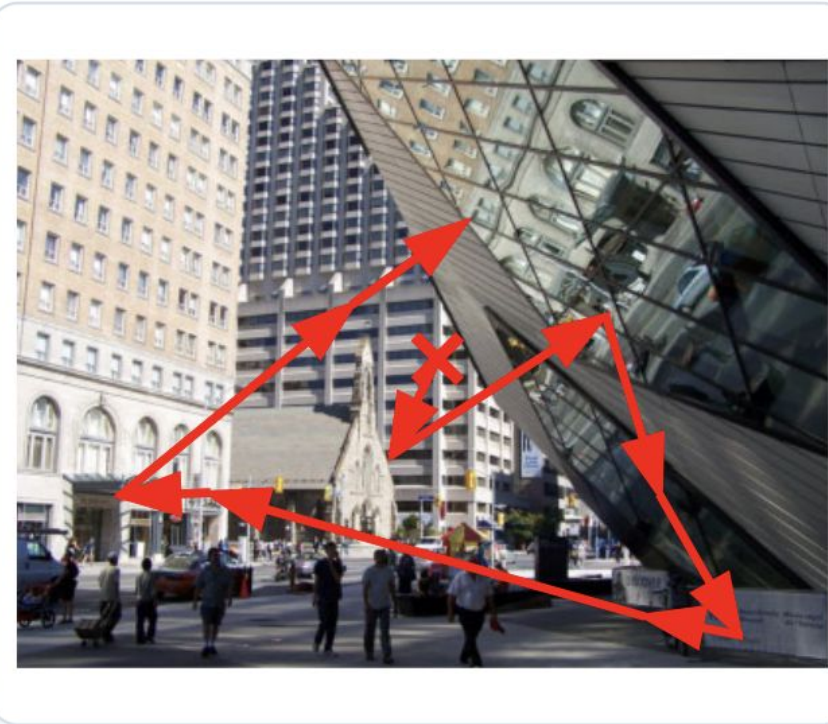
Natural edges show collinear and cocircular relations.

Geisler et al., Sigman et al. (2001)

5 GLOBAL SHAPE ORGANIZES MEDIAL STRUCTURE

Sensitivity peaks follow a shape's medial structure.

Kovács & Julesz (1994)



Research Gap

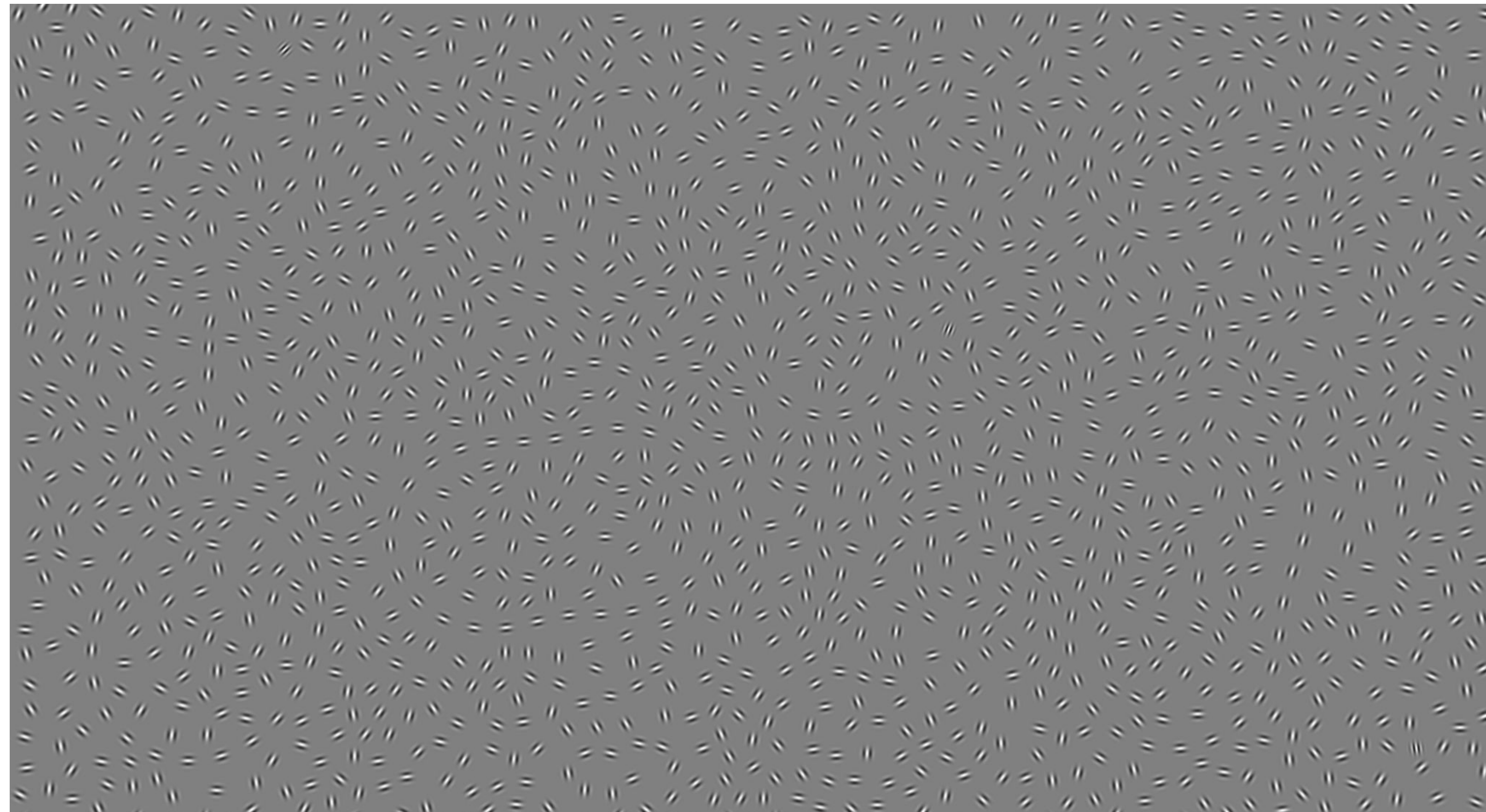
- Natural scenes contain potential confounds, such as object-based attention and semantic information.
- It remains unclear whether eye movements are directly guided toward the stimulus-defined center of mass in the absence of object-level and semantic information.

Hypothesis

Actual fixations will sample higher-LCI locations than would be expected from the observer's general scanpath tendencies alone.

2 TASK & STIMULUS GENERATION

Participant-facing active search



Task

8-s search display

How Many Targets Did You Find? (Press 0-3)

High Spatial Frequency Gabor Patch

LCI Computation

$$C_i(x, r) = O_i(x)D_i(x, r)$$

$$S(x, r) = \sum_{i=1}^N C_i(x, r)$$

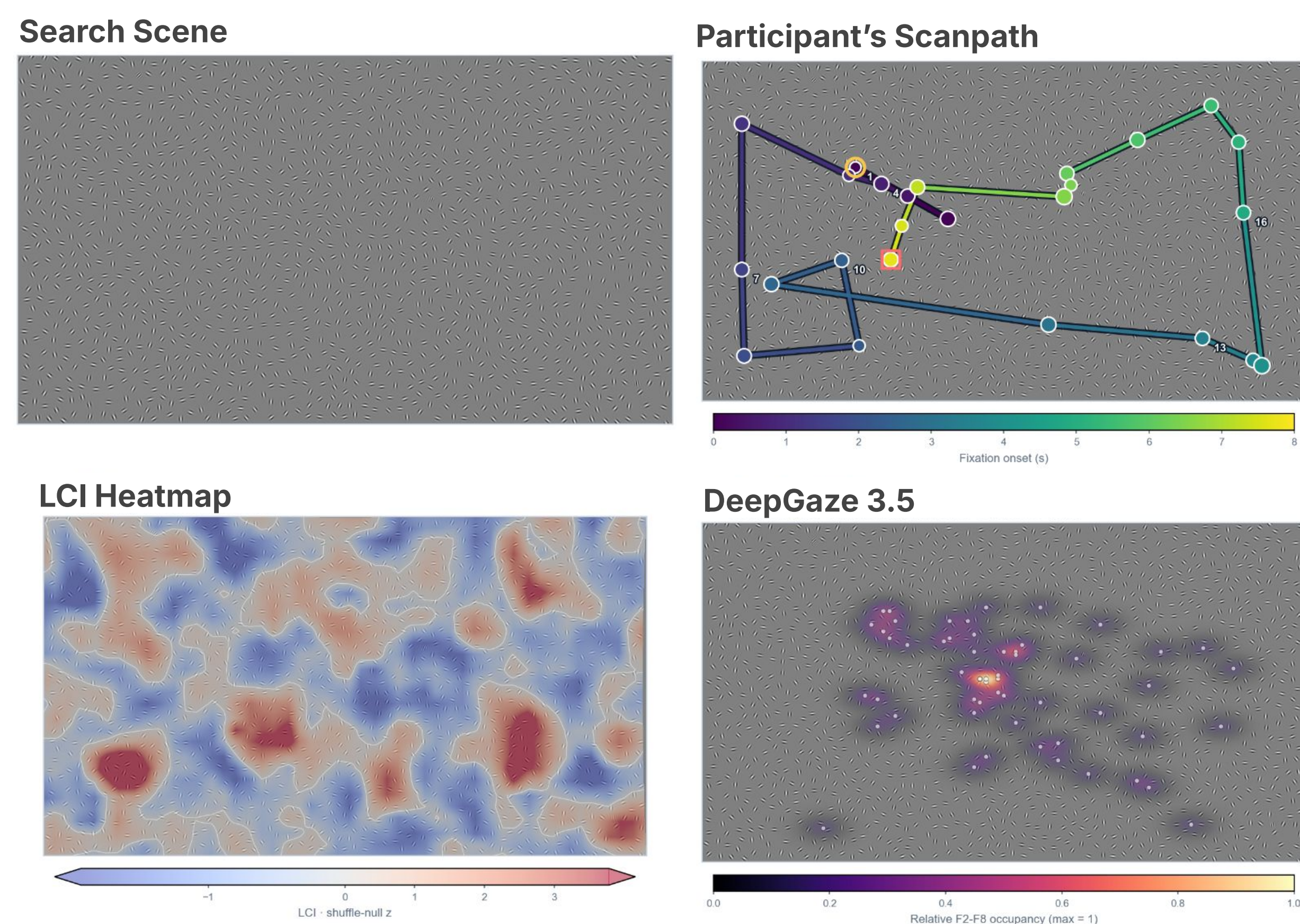
$$S(x, r) = \sum_{i=1}^N \exp \left[-\frac{\Delta \theta_i(x)^2}{2\sigma_\theta^2} - \frac{(d_i(x) - r)^2}{2\sigma_d^2} \right]$$

$$LCI_{hard-max}^{obs}(x) = \max_{r \in \{2.00, 2.25, 2.50, 2.75, 3.00\}^\circ} S(x, r)$$

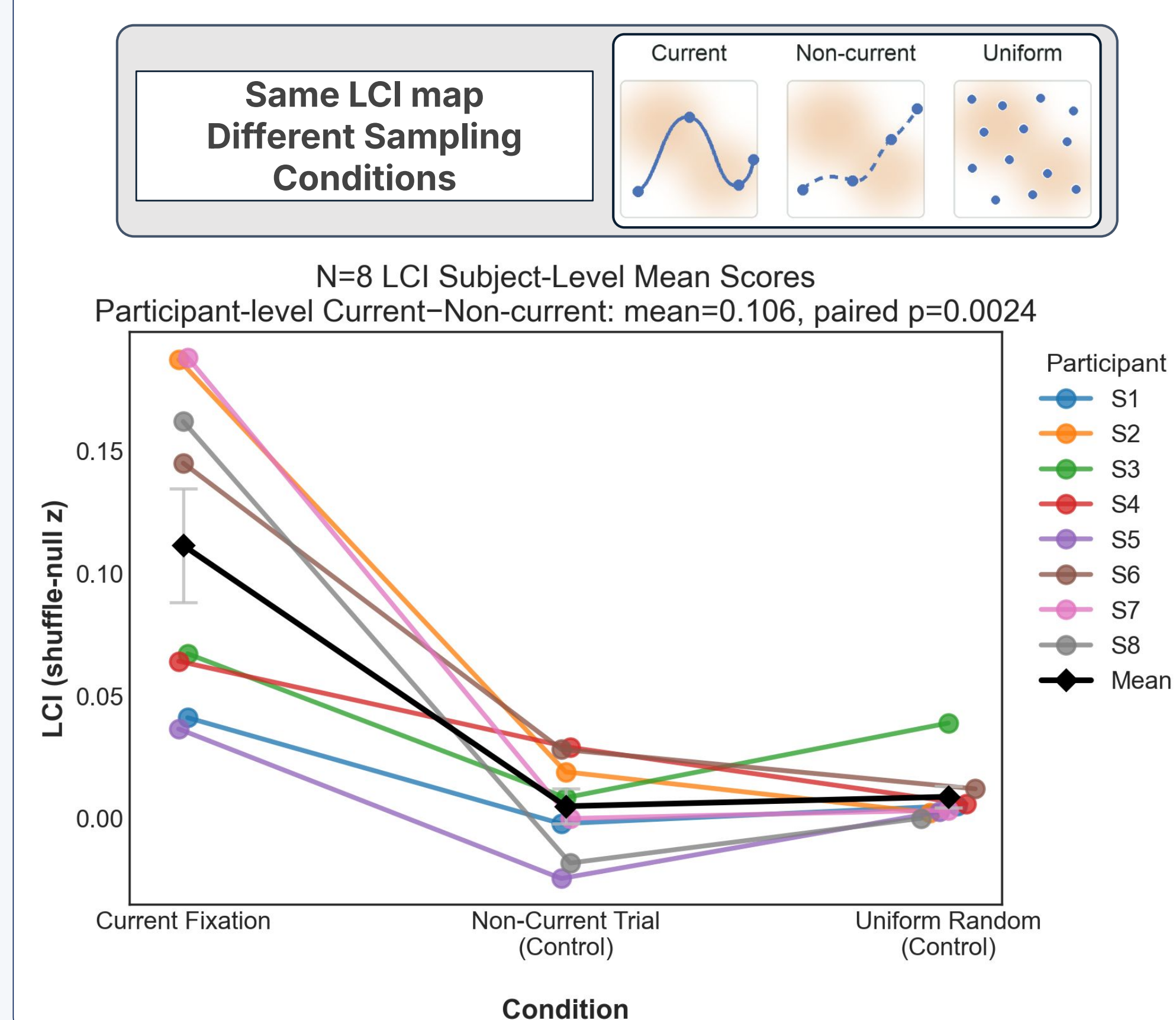
$$LCI(x) = \frac{LCI_{hard-max}^{obs}(x) - \mu(LCI_{hard-max}^{shuffle}(x))}{\sigma(LCI_{hard-max}^{shuffle}(x))}$$

3 MAIN RESULTS

One Trial, Four Representations



Actual fixations sampled higher LCI



4 CONCLUSION & NEXT STEPS

Conclusion

Human gaze selectively samples locations associated with concentric structure in the scene.

1 Candidate-center representation.

LCI converts distributed local orientations into a continuous map of possible object centers.

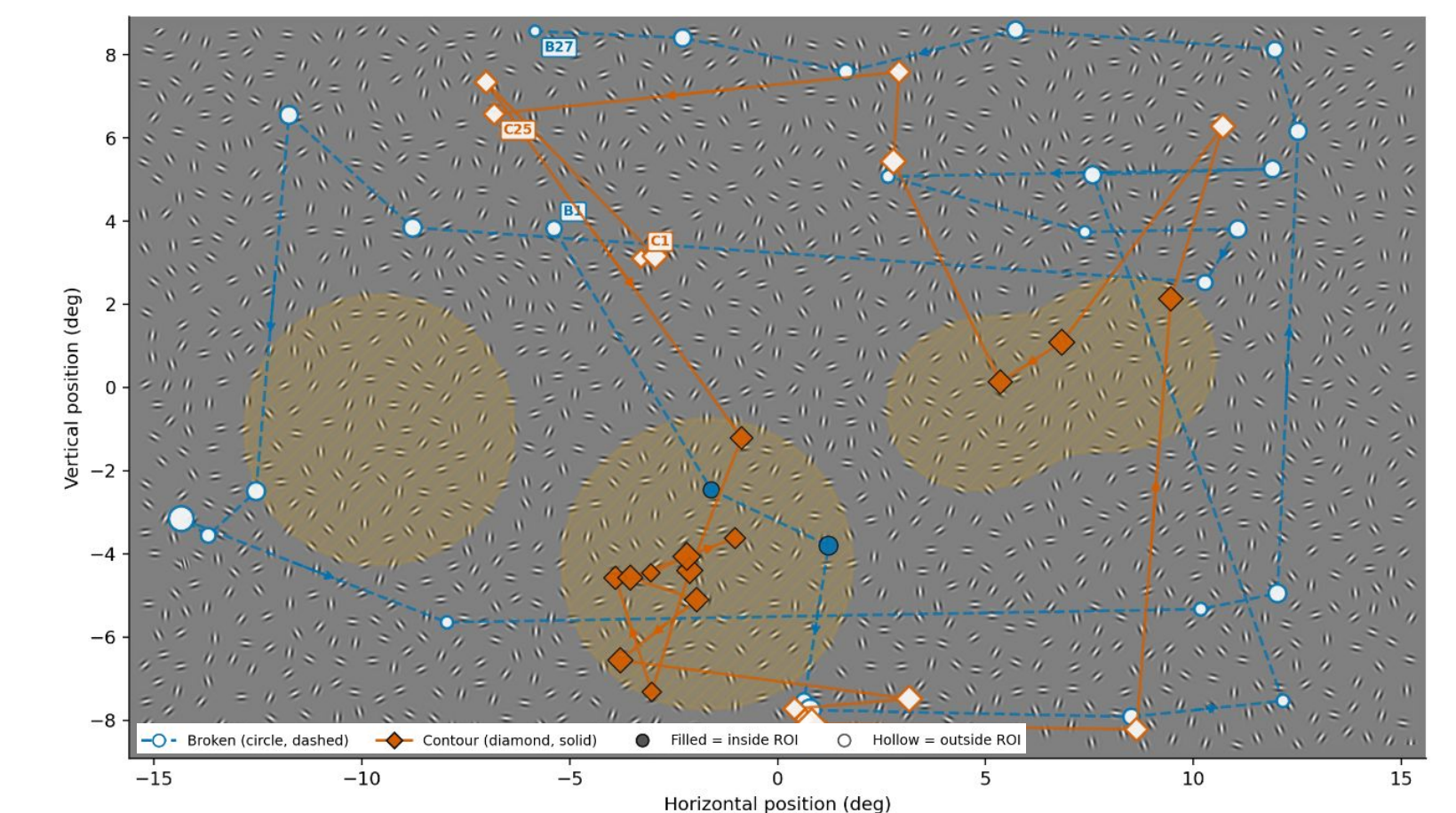
2 Positive gaze-scene alignment.

Actual fixations sampled higher LCI than control.

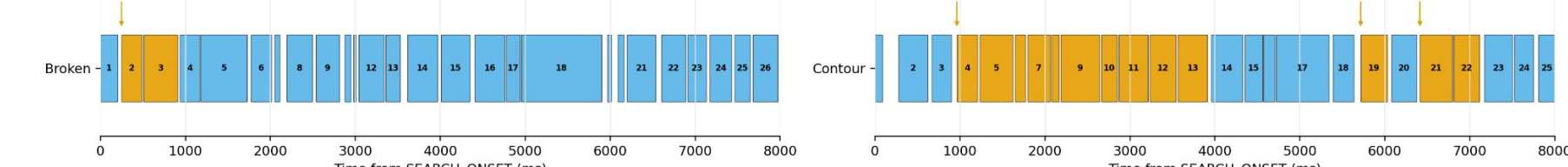
Next steps

Matched contour-broken scanpath comparison

How do scanpaths differ between trials with/ without contours?

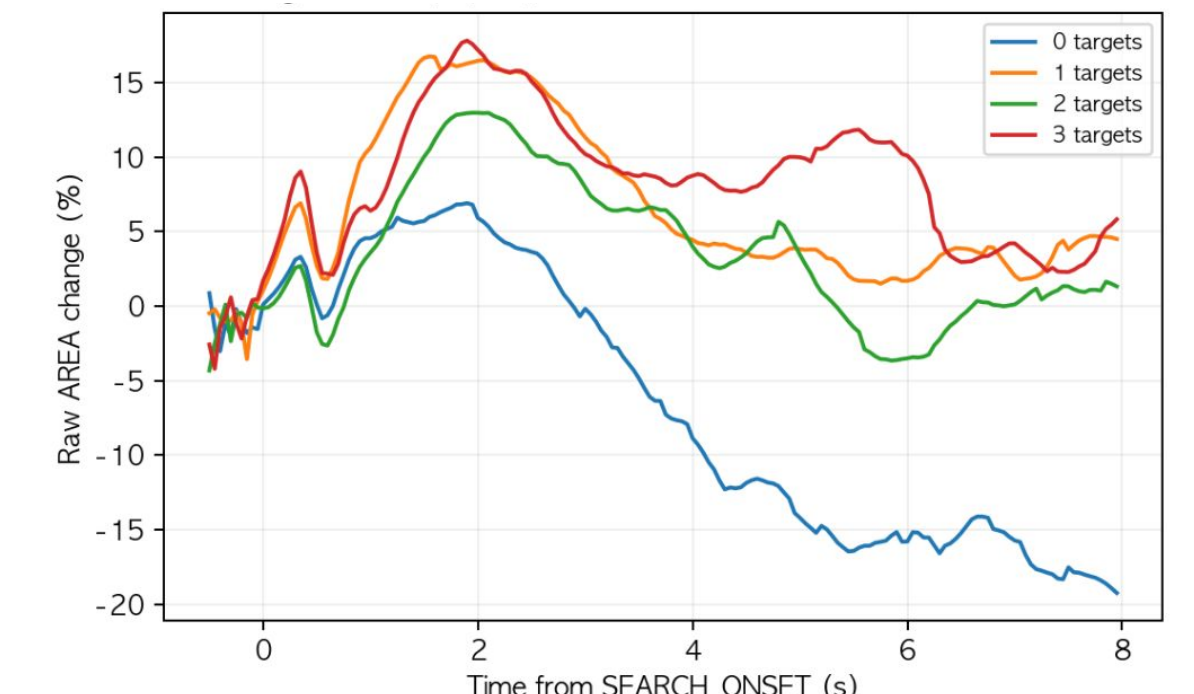


Once gaze enters the contour ROI, does fixation remain there longer?

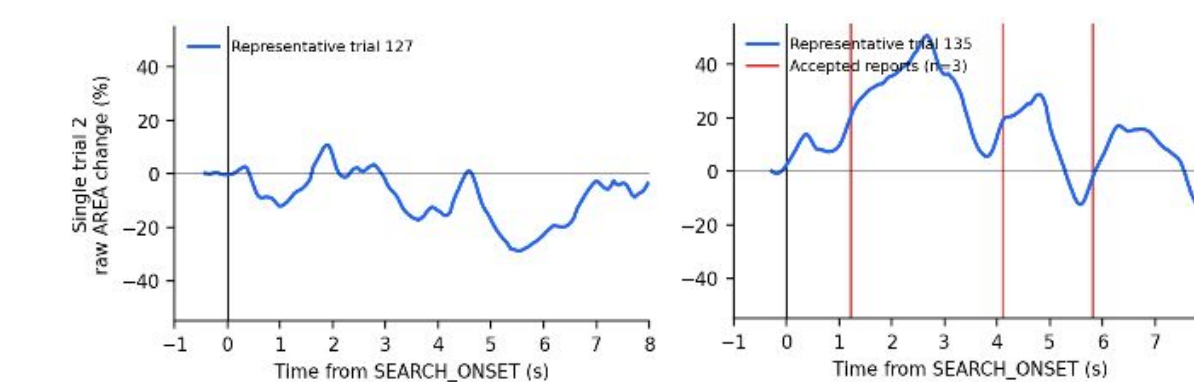


Pupil Dynamics

How does the pupil time course differ between trials with different number of targets?

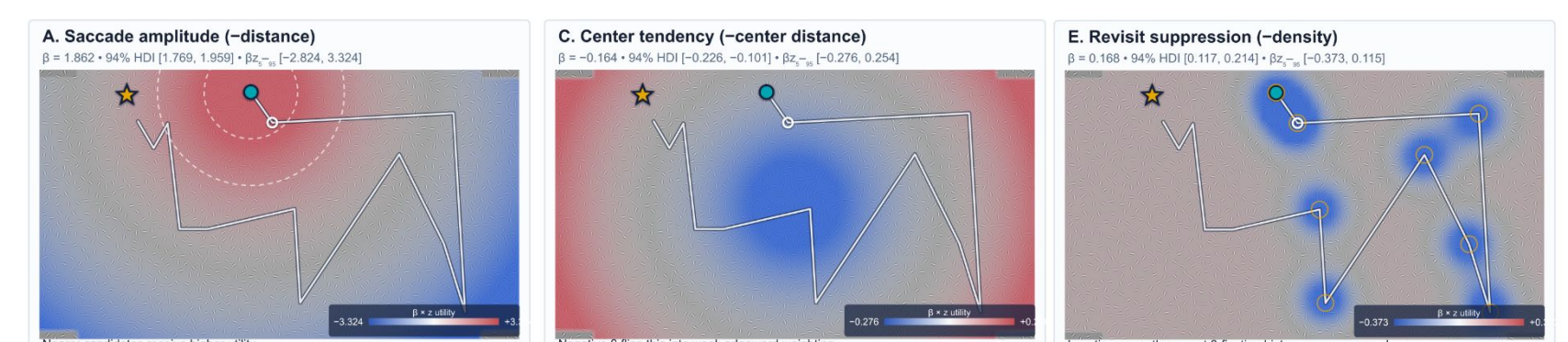


Does pupil size change when a target is detected?



Bayesian Generative Modeling

What posterior distributions over scene-based and gaze-history mechanisms explain fixation selection?



Acknowledgements

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